

ZERO2HERO Cohort 2025

6-Month Roadmap for College Freshers ☐

Daily Task: Watch & solve problems from 'DSA in 90 Days' YouTube playlist

Week 1

- Figma basics: Frame, layout, components, prototyping
- Start designing simple UI screens (landing page, signup)
- Understand importance of design in development

Week 2

- GitHub basics: clone, push, pull, branch
- Create GitHub profile with pinned projects
- Resume writing: tools like Rezi/Overleaf
- LinkedIn optimization: headline, featured, projects

Week 3

- Computer fundamentals: Binary, logic gates, flowcharts
- Choose primary programming language: Java / JS / Python

Weeks 4–5

- Syntax, control flow, functions
- Loops, recursion, arrays, strings
- OOPs: classes, objects, inheritance, polymorphism

Week 6

- Java Generics (List, Map<K,V>)
- Type safety and wildcard types
- Use cases in collections framework

Week 7

- Threads and Runnable interface
- Thread lifecycle and synchronization
- Deadlocks and wait/notify

Weeks 8–9

- OS: processes, memory, scheduling
- Networking: TCP/IP, DNS, HTTP
- DBMS: ACID, ERD, SQL vs NoSQL
- PostgreSQL + MongoDB: schema and queries

Week 10

- Linux terminal, file operations, permissions
- Bash scripting: if, loops, cron jobs

Week 11

- HTML: semantic tags, forms
- CSS: box model, flexbox, grid
- Responsive UI with TailwindCSS

Weeks 12–13

- React components, props, state
- Next.js: routing, pages, API routes
- Tailwind integration
- Build blog or portfolio UI

Week 14

- REST APIs with Node.js and NestJS
- Routing, controllers, DTOs
- Database integration

Week 15

- Turborepo or Nx basics
- Organize frontend + backend in monorepo
- Use shared utilities or types

Weeks 16–17

- LLD: design patterns (Singleton, Factory, Strategy)
- Projects: BookMyShow, URL shortener
- UML diagrams, interface design

Weeks 18–19

- Distributed systems: load balancing, caching, CAP theorem
- Design Instagram/WhatsApp-like systems
- Draw system design diagrams

Week 20

- Build AI-integrated project (image generator, chatbot)
- Use prompt engineering and simple AI APIs

Weeks 21–22

- Dockerize backend services
- Kubernetes: pods, services, deployments
- Deploy on AWS EC2 / Vercel / Render

Weeks 23–24

- Create project demo videos
- Share on LinkedIn with GitHub links
- Engage with mentors and community

Weeks 25–26

- Polish resume, LinkedIn, portfolio
- Set up Notion portfolio or Devfolio
- Apply for internships and referrals